

PIN MAPPING — TEDAC DIPRE_HID

Board: EC Buying FK723M1-ZGT6 V1.0 | MCU: STM32H723ZGT6 (LQFP144) | USB HID Joystick for DCS World

Note: Board pin labels use abbreviated format (e.g.: **E0** = PE0, **B7** = PB7, **A6** = PA6). Standalone **G** labels are GND pins. Buttons: one terminal to **GND**, the other to the GPIO pin (internal pull-up enabled, no external resistor needed). Potentiometers: **3V3** → **end terminal** | **ADC Pin** → **wiper** | **GND** → **end terminal**.

ANALOG AXES — ADC1 (6 channels, 16-bit, circular DMA)

Axis	STM32 Pin	Board Label	ADC Channel	HID Axis	TEDAC Function
1	PA6	A6	ADC1_IN3	X	Axis X
2	PA7	A7	ADC1_IN7	Y	Axis Y
3	PA2	A2	ADC1_IN14	Z	Axis Z
4	PA3	A3	ADC1_IN15	Rx	Axis Rx
5	PC0	C0	ADC1_IN10	Ry	Axis Ry
6	PC1	C1	ADC1_IN11	Rz	Axis Rz

Potentiometer wiring: Pin 1 (end) → **GND** | Pin 2 (wiper) → **ADC Pin** | Pin 3 (end) → **3V3**

USB OTG HS (Full-Speed mode)

STM32 Pin	Board Label	Function
PA11	A11	USB D- (routed to on-board USB connector)
PA12	A12	USB D+ (routed to on-board USB connector)

Do not connect anything externally to these pins. Use the on-board micro-USB / USB-C connector.

TDU BUTTONS — Tactical Display Unit (21 buttons)

BTN #	STM32 Pin	Board	CubeMX Name	TEDAC Function
01	PE0	E0	BTN_01_TDU_TAD	TDU TAD
02	PE1	E1	BTN_02_TDU_FCR	TDU FCR
03	PE2	E2	BTN_03_TDU_PNV	TDU PNV
04	PE3	E3	BTN_04_TDU_GS	TDU GS
05	PE4	E4	BTN_05_TDU_AZ	TDU AZ
06	PE5	E5	BTN_06_TDU_SW6	TDU SW6
07	PE6	E6	BTN_07_TDU_SW7	TDU SW7
08	PE7	E7	BTN_08_TDU_LMC	TDU LMC
09	PE8	E8	BTN_09_TDU_CAGE	TDU CAGE
10	PE9	E9	BTN_10_TDU_RF_UP	TDU RF UP
11	PE10	E10	BTN_11_TDU_RF_DOWN	TDU RF DOWN
12	PE11	E11	BTN_12_TDU_EL_UP	TDU EL UP
13	PE12	E12	BTN_13_TDU_EL_DOWN	TDU EL DOWN
14	PE13	E13	BTN_14_TDU_SYM_UP	TDU SYM UP
15	PE14	E14	BTN_15_TDU_SYM_DOWN	TDU SYM DOWN
16	PE15	E15	BTN_16_TDU_BRT_UP	TDU BRT UP
17	PD8	D8	BTN_17_TDU_BRT_DOWN	TDU BRT DOWN
18	PD9	D9	BTN_18_TDU_COM_UP	TDU COM UP
19	PD10	D10	BTN_19_TDU_COM_DOWN	TDU COM DOWN

BTN #	STM32 Pin	Board	CubeMX Name	TEDAC Function
20	PD11	D11	BTN_20_TDU_DAY	TDU DAY
21	PD12	D12	BTN_21_TDU_NT	TDU NT

■ **LHG BUTTONS — Left Hand Grip (26 buttons)**

BTN #	STM32 Pin	Board	CubeMX Name	TEDAC Function
22	PD0	D0	BTN_22_LHG_SW1	LHG SW1
23	PD1	D1	BTN_23_LHG_SW2	LHG SW2
24	PD2	D2	BTN_24_LHG_SW3	LHG SW3
25	PD3	D3	BTN_25_LHG_SW4	LHG SW4
26	PD4	D4	BTN_26_LHG_SW5	LHG SW5
27	PD5	D5	BTN_27_LHG_HAT1_UP	LHG HAT1 UP
28	PD6	D6	BTN_28_LHG_HAT1_DOWN	LHG HAT1 DOWN
29	PD7	D7	BTN_29_LHG_HAT1_LEFT	LHG HAT1 LEFT
30	PB0	B0	BTN_30_LHG_HAT1_RIGHT	LHG HAT1 RIGHT
31	PB1	B1	BTN_31_LHG_HAT2_UP	LHG HAT2 UP
32	PB2	B2	BTN_32_LHG_HAT2_DOWN	LHG HAT2 DOWN
33	PB3	B3	BTN_33_LHG_HAT2_LEFT	LHG HAT2 LEFT
34	PB4	B4	BTN_34_LHG_HAT2_RIGHT	LHG HAT2 RIGHT
35	PB5	B5	BTN_35_LHG_HAT3_UP	LHG HAT3 UP
36	PB6	B6	BTN_36_LHG_HAT3_DOWN	LHG HAT3 DOWN
37	PB7	B7	BTN_37_LHG_HAT3_LEFT	LHG HAT3 LEFT
38	PB8	B8	BTN_38_LHG_HAT3_RIGHT	LHG HAT3 RIGHT
39	PB9	B9	BTN_39_LHG_TEMP1_A	LHG TEMP1 A
40	PB10	B10	BTN_40_LHG_TEMP1_B	LHG TEMP1 B
41	PB11	B11	BTN_41_LHG_TEMP2_A	LHG TEMP2 A
42	PB12	B12	BTN_42_LHG_TEMP2_B	LHG TEMP2 B
43	PB13	B13	BTN_43_LHG_TEMP3_A	LHG TEMP3 A
44	PC4	C4	BTN_44_LHG_TEMP3_B	LHG TEMP3 B
45	PC5	C5	BTN_45_LHG_3POS_A	LHG 3POS A
46	PC6	C6	BTN_46_LHG_3POS_B	LHG 3POS B
47	PC7	C7	BTN_47_LHG_PUSH	LHG PUSH

■ **RHG BUTTONS — Right Hand Grip (22 buttons)**

BTN #	STM32 Pin	Board	CubeMX Name	TEDAC Function
48	PC8	C8	BTN_48_RHG_SW1	RHG SW1
49	PC9	C9	BTN_49_RHG_SW2	RHG SW2
50	PC10	C10	BTN_50_RHG_SW3	RHG SW3
51	PC11	C11	BTN_51_RHG_SW4	RHG SW4
52	PC12	C12	BTN_52_RHG_SW5	RHG SW5
53	PC13	C13	BTN_53_RHG_SW6	RHG SW6
54	PF0	F0	BTN_54_RHG_SW7	RHG SW7
55	PF1	F1	BTN_55_RHG_HAT1_UP	RHG HAT1 UP
56	PF2	F2	BTN_56_RHG_HAT1_DOWN	RHG HAT1 DOWN
57	PF3	F3	BTN_57_RHG_HAT1_LEFT	RHG HAT1 LEFT
58	PF4	F4	BTN_58_RHG_HAT1_RIGHT	RHG HAT1 RIGHT
59	PF5	F5	BTN_59_RHG_HAT2_UP	RHG HAT2 UP
60	PF6	F6	BTN_60_RHG_HAT2_DOWN	RHG HAT2 DOWN
61	PG6	G6	BTN_61_RHG_HAT2_LEFT	RHG HAT2 LEFT
62	PG7	G7	BTN_62_RHG_HAT2_RIGHT	RHG HAT2 RIGHT
63	PG8	G8	BTN_63_RHG_TEMP_A	RHG TEMP A
64	PG9	G9	BTN_64_RHG_TEMP_B	RHG TEMP B

BTN #	STM32 Pin	Board	CubeMX Name	TEDAC Function
65	PG10	G10	BTN_65_RHG_3POS1_A	RHG 3POS1 A
66	PG11	G11	BTN_66_RHG_3POS1_B	RHG 3POS1 B
67	PG12	G12	BTN_67_RHG_3POS2_A	RHG 3POS2 A
68	PG13	G13	BTN_68_RHG_3POS2_B	RHG 3POS2 B
69	PG14	G14	BTN_69_RHG_PUSH	RHG PUSH

RESERVED PINS — Do Not Connect

STM32 Pin	Label	Function
PA11	A11	USB OTG HS D- (on-board USB connector)
PA12	A12	USB OTG HS D+ (on-board USB connector)
PH0	(HSE)	25 MHz Crystal (on-board)
PH1	(HSE)	25 MHz Crystal (on-board)
PA13	(SWDIO)	SWD Debug — Data I/O
PA14	(SWCLK)	SWD Debug — Clock

POWER SUPPLY & GND

Board Label	Description
3V3	Regulated 3.3V output — use to power potentiometers and sensors
5V	5V input/output from USB — main power supply
G	GND — multiple pins on both rows (use for buttons and potentiometers)
Vref	ADC reference voltage — connect to 3.3V or leave unconnected
BAT	VBAT — RTC backup battery, not used in this project

GPIO PORT SUMMARY

Port	Pins Used	Count	Usage
PA	PA2, PA3, PA6, PA7, PA11, PA12	6	4 ADC + 2 USB
PB	PB0 — PB13	14	14 buttons
PC	PC0, PC1, PC4 — PC13	12	2 ADC + 10 buttons
PD	PD0 — PD12	13	13 buttons
PE	PE0 — PE15	16	16 buttons
PF	PF0 — PF6	7	7 buttons
PG	PG6 — PG14	9	9 buttons

TOTAL: 77 pins used	6 ADC + 2 USB + 69 Buttons
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QUICK WIRING DIAGRAMS

Potentiometer (x6):

- 3V3 End terminal 1
- ADC Pin Wiper (center tap) → A6, A7, A2, A3, C0, C1
- GND End terminal 2

Button / Switch (x69):

- GPIO Pin Button terminal 1
- GND Button terminal 2

Internal pull-up enabled. No external resistor needed.
Button pressed = contact to GND = GPIO reads LOW = bit = 1 in HID report.